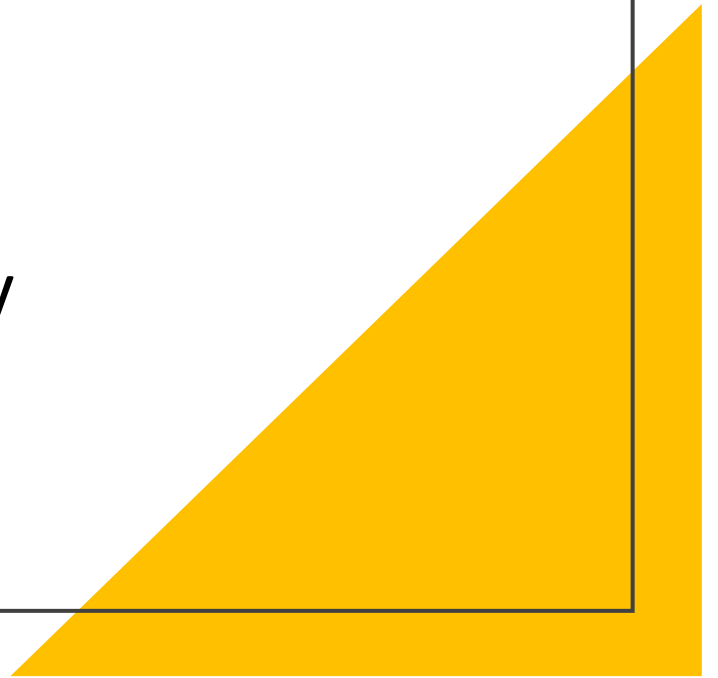


Sociotechnical Security

CS 8803/4803

Professor Veronica Rivera

Lecture 2 – Methods in sociotechnical security



Last time

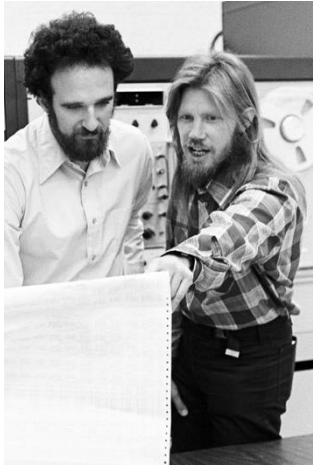
- Looked at examples of computer system failures and humans' role in them
- Course logistics
- Why solving these problems is challenging using traditional security methods (math)
- You read “How to read a paper”

Today: Sociotechnical security toolbox

- History of sociotechnical security in the context of major internet events
- Survey the tools we have at our disposal to try to understand and solve these problems
 - Focus on human-centered and empirical techniques
 - Very very high-level overview. This is not a methods course.

A (highly condensed and US-centric) history of
sociotechnical security

1980s-1990s: Online services & modern encryption



1976: Diffie and Hellman
publish public-key crypto



1991: Cubby, Inc v.
CompuServe



1995: Stratton Oakmont v.
Prodigy



1996: Section 230 + CDA



1999: First usable s&p papers

2000s: Security problems arose as people came online



2003: CAN-SPAM Act



2004: Facebook launches



2005: SOUPS conference
starts



2005: YouTube launches

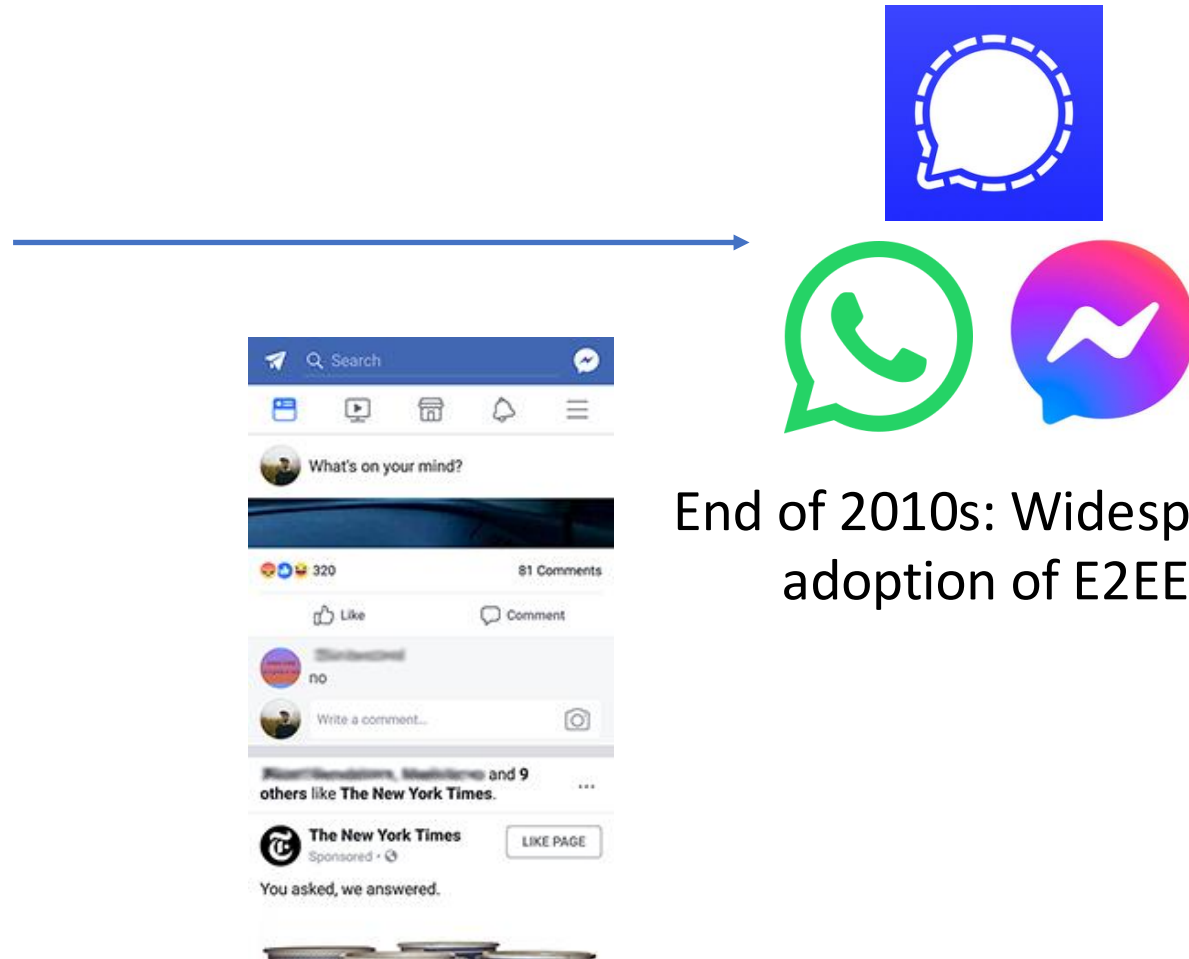


2006: Twitter launches

2010s: security becomes universal, social harms persist



2013: Snowden leaks



End of 2010s: Widespread adoption of E2EE

2019: Christchurch terrorist attack

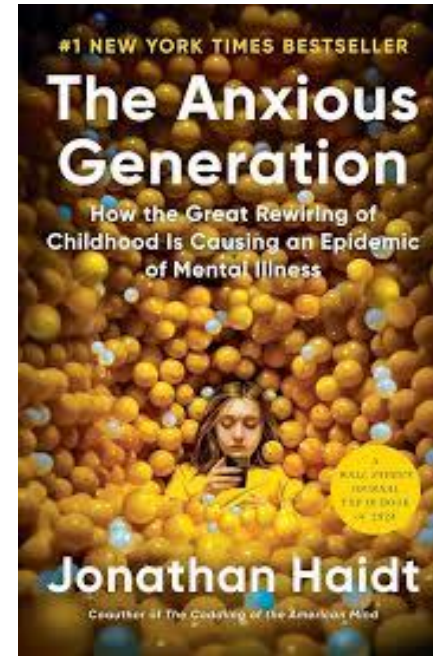
2020s: Tech is widespread & mediates social harm



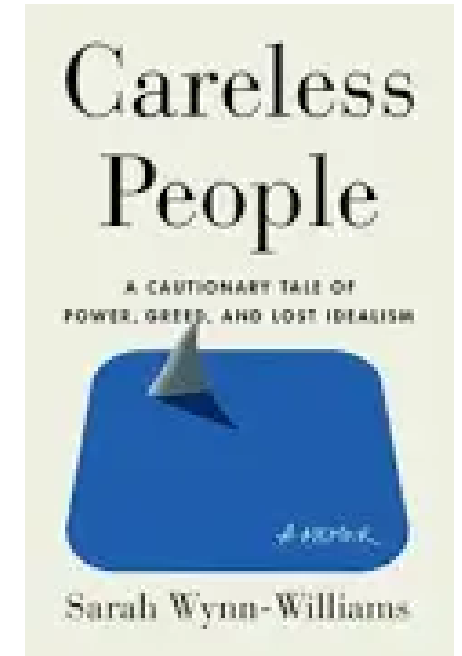
2021: Insurrection of US Capitol



2022: OpenAI releases GPT-3



2024: The Anxious Generation



2025: Careless People

Usable security and privacy remain important: new systems like private search engines, private reputation systems, and passkeys will impact users.

So what tools do we have to solve these problems?

Sociotechnical security toolbox

This is by far not an exhaustive list!!

- Qualitative studies
- Survey studies
- Usability studies
- Measuremet studies



Qualitative methods

Qualitative methods

Not an exhaustive list!

- Interviews
- Focus groups
- Diary studies
- Participatory design
- Ethnography/participant observation
- Contextual inquiry
- Content and discourse analysis

What are qualitative methods good for

- Establish a threat model for a new kind of harm
- Understand users' constraints in adopting protective tools
- Identify tensions between different user values
- Give voice to populations underrepresented in existing security research or design

What interviews are good for

- Explain a quantitative finding you don't yet understand (e.g., why a metric spiked or a feature went unused)
- Generate hypotheses or categories to test with a larger, quantitative study later
 - Especially when there is not much work on a given problem yet
- Identify what interventions you want to design, implement, or test

Example: Establishing a threat model labor abuse reporting in digital whisper networks

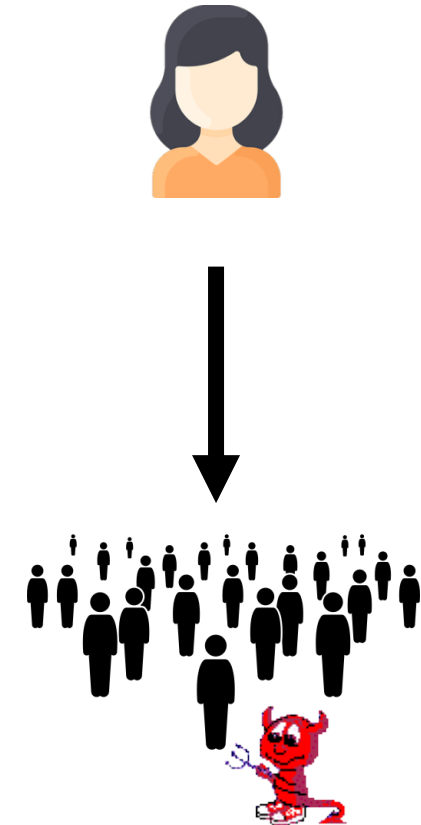
Goals, Risks, and Safety Practices in Online Labor Abuse Disclosures

Veronica A. Rivera^{§†*}, Tracy Li[†], Alex Ozdemir^{§*},
Catherine Han[†], Zakir Durumeric[†], Elissa M. Redmiles[‡]

[§]*Max Planck Institute for Security and Privacy* [†]*Stanford University*

^{*}*Georgia Institute of Technology* [‡]*Georgetown University*

- How are victims of labor abuse navigating safety considerations when disclosing online?
- What does this threat model look like?
- What kinds of things should we consider when designing tools to support these disclosures?
- Interviews with 17 participants who have disclosed labor abuse online



https://vriviera2017.github.io/docs/SOUPS26_DWN.pdf

Research questions

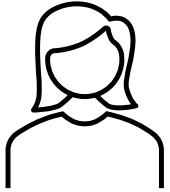
Structure: What types of technologies do participants use to disclose their experiences?

Goals: What goals do participants have when sharing their experiences? How do these shape which technologies they use?

Risks: What risks do participants experience or perceive when using DWNs? How do these relate to the disclosure and/or the tech used?

Mitigations: What decisions do participants make during the process of disclosure? How do they intersect with disclosure goals?

Tensions: obfuscation & anonymity

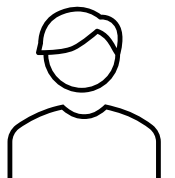


“I try not to share geographically locating information, identifiable information, or names and places”. (P8)

Obfuscation & anonymity reduce retaliation ...

... but increase other harms.

“You don’t have that level of detail about these things unless you are truly going through them”. (P17)



How To Do Interviews

- Draft your protocol
 - with ~10 questions for a 45 minute interview
- Practice
 - Get feedback on your questions to make sure they are interpreted correctly
- Recruit
 - Social media, snowball sampling, flyering
- Interview
 - Asking participants if you can audio record will make analysis easier
- Analyze
 - Transcribe your interviews right away
 - Analyze using a process called “coding” (e.g., thematic analysis or grounded theory)
 - DO NOT PUT YOUR TRANSCRIPTS INTO LLMs WITHOUT PARTICIPANT CONSENT

Recruitment

- Identify a user group relevant to your research question
- Recruitment survey
 - Informed consent
 - Screen for particular demographics/characteristics
- Consider how your choice of where to recruit influences results
- Aim for a representative sample
 - If your sample is difficult to recruit snowball sampling can be helpful
- In qualitative research, you often recruit and analyze simultaneously

Types of questions to ask

- Ask open-ended questions rather than questions that lead to a short, direct response
- Probe participants to tell you more
 - Probing questions:
 - Interesting...tell me more...
 - Can you expand on that?
 - Can you give me a bit more detail?
 - ...go on...
 - What happened next?
 - Do you have any other examples?
 - Can you describe a time when x happened?

Example questions

Good questions

Threats to Sharing.

1. Thinking back to the times you have shared stories and information about your negative experiences online, were you worried about anything bad happening when you posted?
 - a. If yes:
 - i. What bad things were you worried about?
 - ii. What decisions and tradeoffs did you consider when sharing with this group?
 - iii. What things about the group made you feel safe/unsafe when sharing?
3. What do you see as the threats to the goals/purpose/existence of the digital reporting networks you're a part of?

Bad questions

Leading

Impossible

- Beyond expertise
- Functional fixedness

Use very very rarely

- Closed
 - Binary
 - Counts / likerts
- Hypothetical

Example:

- What would you like in a tool for [X]?
- Would you like to do less work in this class?

Limitations of interviews

- You can't generalize beyond the group you talked to
 - Sample size is too small
- Hard to compare across participants meaningfully
- Doesn't capture behavior directly – captures *accounts* of behavior
- Coding subjectivity, even with multiple coders
- Various biases:
 - Social desirability
 - Recall bias
- Time-intensive & hard to anticipate/control data quality

Surveys

What are survey methods good for

- How many or how common
- Whether groups differ
- Correlations between variables
- Generalizing beyond your
- Benchmarking against standardized instruments
- Quantifying something qualitative work already identified

Example: Measuring attitudes around acceptability of deepfake creation & sharing

“Violation of my body:”

Perceptions of AI-generated non-consensual (intimate) imagery

Natalie Grace Brigham
University of Washington

Miranda Wei
University of Washington

Tadayoshi Kohno
University of Washington

Elissa M. Redmiles
Georgetown University



- Survey of 315 people in the US on their views surrounding hypothetical non-consensual creation of deepfakes depicting them
- Measured differences in attitudes around creating, sharing, and seeking out AI-generated sexual & non-sexual deepfakes
 - Measured relationship between attitudes and different variables (e.g., gender, relationship to creator) <https://www.usenix.org/system/files/soups2024-brigham.pdf>

RQs/study goals

How do attitudes towards non-consensual synthetic imagery differ depending on:

RQ1) **what** is depicted: sexual vs. non-sexual acts

RQ2) **who** is creating the media and for what **purpose**?

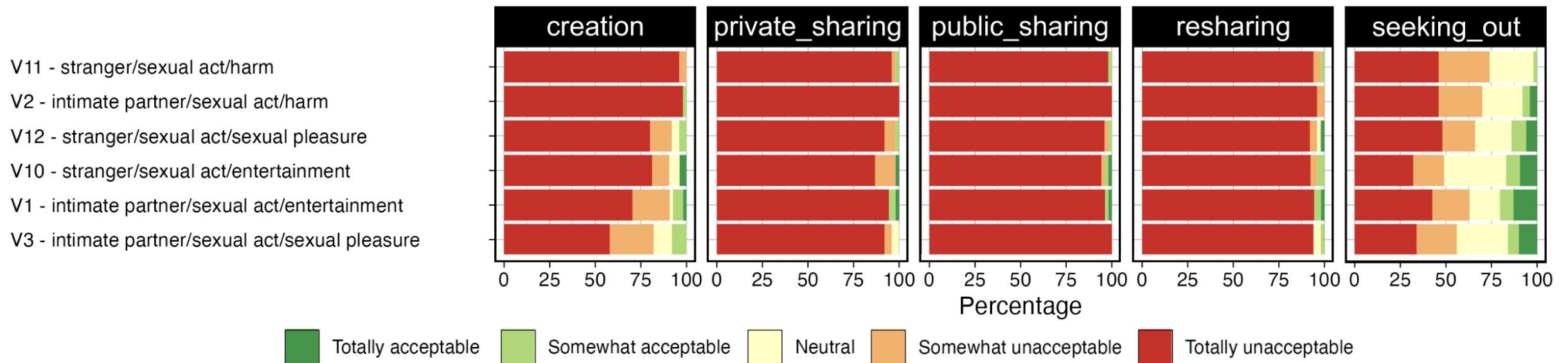
RQ3) attitudes towards **sexual (a) consent** and **(b) non-synthetic content**?

RQ4) **gender**?

Setup: vignettes

Creator	Action	Intent	Media Status
Stranger Intimate partner	Performing a sexual act	To harm	Creation
	Saying something	To entertain	Public/private sharing
	Playing a sport	For sexual pleasure	Resharing Seeking out

Example results



Respondents' perceptions of acceptability across vignettes involving AIG-NCII; each vignette is defined by the creator / action / intent.

Results: hypothesis tests

- Observation: more people are ok with **creation** than **resharing**.
- Question: is that a real pattern, or noise in the data?
- Answer technique: hypothesis testing
 - Roughly: “Given the noise we observed, what is the probability of seeing **creation** > **resharing**, if they were actually equal?” $\leftarrow$ “p-value”
 - Here it is 0.001---not likely, so the pattern is real.

ble 6 in Appendix B for full results): Across scenarios and controlling for within-subject variation we observe that private_sharing (OR = 0.47, $p < 0.001$), public_sharing (OR = 0.26, $p < 0.001$), and resharing (OR = 0.42, $p < 0.001$) are significantly less acceptable than creation (the reference level). seeking_out (OR = 5.43, $p < 0.001$) is significantly more acceptable than creation.

Results: more tests and correlations

		creation	private_sharing	public_sharing	resharing	seeking_out
Controlled IVs	creator (Intimate partner)	3.24*** [2.23, 4.71]	1.69* [1.13, 2.55]	1.47 [0.9, 2.4]	1.00 [0.65, 1.53]	1.11 [0.8, 1.53]
	action (Sport)	13.39*** [7.96, 22.52]	34.72*** [16.76, 71.92]	66.61*** [22.75, 195.04]	32.36*** [15.12, 69.25]	7.26*** [4.73, 11.15]
	action (Saying something)	5.44*** [3.27, 9.05]	11.01*** [5.45, 22.23]	19.49*** [6.91, 54.94]	12.47*** [5.92, 26.29]	3.40*** [2.21, 5.22]
	intent (Entertainment)	18.92*** [11.03, 32.46]	11.49*** [6.59, 20.05]	10.57*** [5.39, 20.73]	5.51*** [3.18, 9.56]	4.94*** [3.25, 7.49]
	intent (Other)	7.42*** [4.42, 12.47]			0.92 [0.52, 1.63]	1.37 [0.92, 2.04]

Correlation strength

Hypothesis test result
(** = $p < 0.001$)

How to do surveys (high-level)

- Think about what you are trying to measure
- Identify whether there are existing scales you can use to measure that
- If there are not existing measures, write questions that will help you understand/measure the thing you care about
 - Don't try to write a scale! – Good scales are hard to write. Questions in a scale do not stand alone
- Test your survey both for timing and to make sure your answers are interpreted accurately

Survey best practices

- Choose wording carefully
 - Problem: Different respondents have different interpretation of terms
 - Strategy: pre-test your survey
 - Problem: Respondents are not familiar with technical jargon
 - Strategy: Discuss which terms you think respondents will be comfortable with
 - Problem: Providing too few or too many points may negatively impact interpretability of results
 - Strategy: Use scale lengths between 5 and 10; Veronica usually uses 5
- Avoid double barreled questions
 - Example: “Do you believe that your employer should require you to update your computer and change your password every six months?”

Survey best practices

- The order in which a question is shown (e.g. the questions asked previously) can affect responses
 - Strategy: Randomize the order in which questions are asked to different people to ensure the bias is randomly distributed

Survey best practices

- People underreport socially undesirable behaviors (e.g. drug use) and overreport socially desirable behaviors (e.g. exercise)
 - This is called desirability bias
- Strategies:
 - Ask indirectly rather than about respondent themselves
 - Soften the question with forgiving wording (e.g. “People have many different reasons for choosing to update or not update their phones. Which of the following best describes your behavior...”)
- Ask demographic questions at the end of the survey

Survey best practices

- Requiring respondents to provide answers to every question is unethical
 - Make sure to provide a “don’t know” or “prefer not to answer” option
- Similarly for interviews, let participants know they can choose not to answer a question.
- Make sure surveys are a reasonable length
 - Longer surveys have a lower response/completion rate
 - Include a completion bar to help participants know how much longer they have left
 - Give participants an estimate of how much time the survey will take (based on your pre-testing)
 - Only ask questions that are truly relevant

Survey limitations

- Can't capture why people select the responses they do
- No opportunity to probe further
- Response options may constrain what people express
- Measures stated attitudes/intentions, not actual behavior
- More recent: many survey respondents are using LLMs
 - Recent paper on how to detect and mitigate this:
<https://www.usenix.org/system/files/soups2026-xu.pdf>
- Measure correlation, not causation

Usability studies

What usability studies are good for

- Controlling the factors that may impact the usability of your system
- Observe users using a system, whether online or in-person
- Reveals usability problems that people can't articulate
- Good for identifying concrete fixes
- Ability to replicate your results

Example: Usability study of five 2FA methods

A Usability Study of Five Two-Factor Authentication Methods

Ken Reese, Trevor Smith, Jonathan Dutson, Jonathan Armknecht, Jacob Cameron, Kent Seamons
Brigham Young University



- Two-week between subjects usability study of five 2FA methods with 72 participants, collecting both qualitative and quantitative data
 - Participants used 2FA to log into simulated banking website to complete task
- Within subjects laboratory study to understand how users set up the five 2FA methods on their devices

RQs/study goals

- Studied 5 2FA methods:
 - SMS, TOTP, pre-generated codes, push, and U2F security key
- Compare these five methods within a single application
 - To reduce confounding factors – are usability differences because of apps or because of 2FA?
 - Gather timing data
- Examine setup and day-to-day usability independently to avoid bias

Results: quantitative & qualitative data

Quantitative

- Timing
 - Do people get faster overtime as they become more familiar with the 2FA method?
 - Differences in users' speed across 2FA methods?
- Usability
 - Which method is more usable, using standardized scale from UX research (System Usability Scale or SUS). <https://ixdf.org/literature/article/system-usability-scale>

Qualitative

- Security and convenience
 - Do people feel more secure logging in with a second factor?
- Prior compromise vs. 2FA inconvenience
 - How do 2FA behaviors change after account compromise

Results: quantitative data

Table 1: Repeated measures correlation (rmcorr) between amount of time participating in study versus amount of time to authenticate.

2FA Method	p-value	r	df	95% confidence interval
SMS	0.280	-0.097	124	(-0.269, 0.081)
TOTP	0.586	-0.049	122	(-0.225, 0.129)
Push	0.029	-0.204	113	(-0.374, -0.020)
U2F	<0.003	-0.269	118	(-0.429, -0.093)
Codes	0.426	-0.076	110	(-0.260, 0.113)

Table 2: Authentication Time (seconds), Summary Statistics

Authentication Method	Q1	Median	Mean	Q3
Codes	11.3	17.2	28.0	25.4
Push	8.4	11.8	16.1	17.6
SMS	13.0	16.6	18.5	22.1
TOTP	10.7	15.1	23.9	23.3
U2F	4.5	9.1	13.0	16.3

Table 3: SUS Scores for each Method, Summary Statistics

Authentication Method	Q1	Median	Mean	Q3
Password	87.5	95.0	92.5	98.8
Codes	75.0	80.0	80.2	90.0
Push	72.5	81.3	81.0	92.5
SMS	68.8	75.0	75.0	80.0
TOTP	75.0	88.8	83.1	92.5
U2F	61.9	75.0	73.1	93.1

How to do a usability study

- Need to design your experiment
 - What are you measuring?
 - What factors will you control for?
 - What might be confounding factors?
- May need to build artifacts to carry out your experiment
 - Prototype you want to test
 - Platform that participants will engage with
- As always, test your experiment with others before running it with real participants!

Limitations of usability studies

- Often times conducted under “ideal” lab conditions
 - E.g., in the example, the banking website was artificial
 - Doesn’t capture how people use a tool under everyday conditions
- Observer effect
 - Being watched may change how participants complete a task
- Small sample

Measurement studies

Types of measurement studies

Not an exhaustive list!

- Web/app crawling and analysis
- Network/traffic measurement
- Policy and interface auditing at scale
- Corpus/document analysis at scale

What measurement studies are good for

- Scale beyond what a survey could do
- Reveals actual practice, not stated policy or self-report
- High external validity for the artifacts studied
- Uncovers systemic patterns
- Often reproducible

Example: Measuring AI companion chatbot creation on Character.AI

- Measurement study investigating why users create and engage with chatbots on Character.AI
- Collected 5,761,412 user-created text inputs that define character on the platform and analyzed a subset of these
- Also analyzed ~2k posts from r/characterai to understand how users interact with AI companions

Caught in a Mafia Romance: How Users Explore Intimate Narratives with Chatbots

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<https://dl.acm.org/doi/pdf/10.1145/3772318.3790806>

RQs/study goals

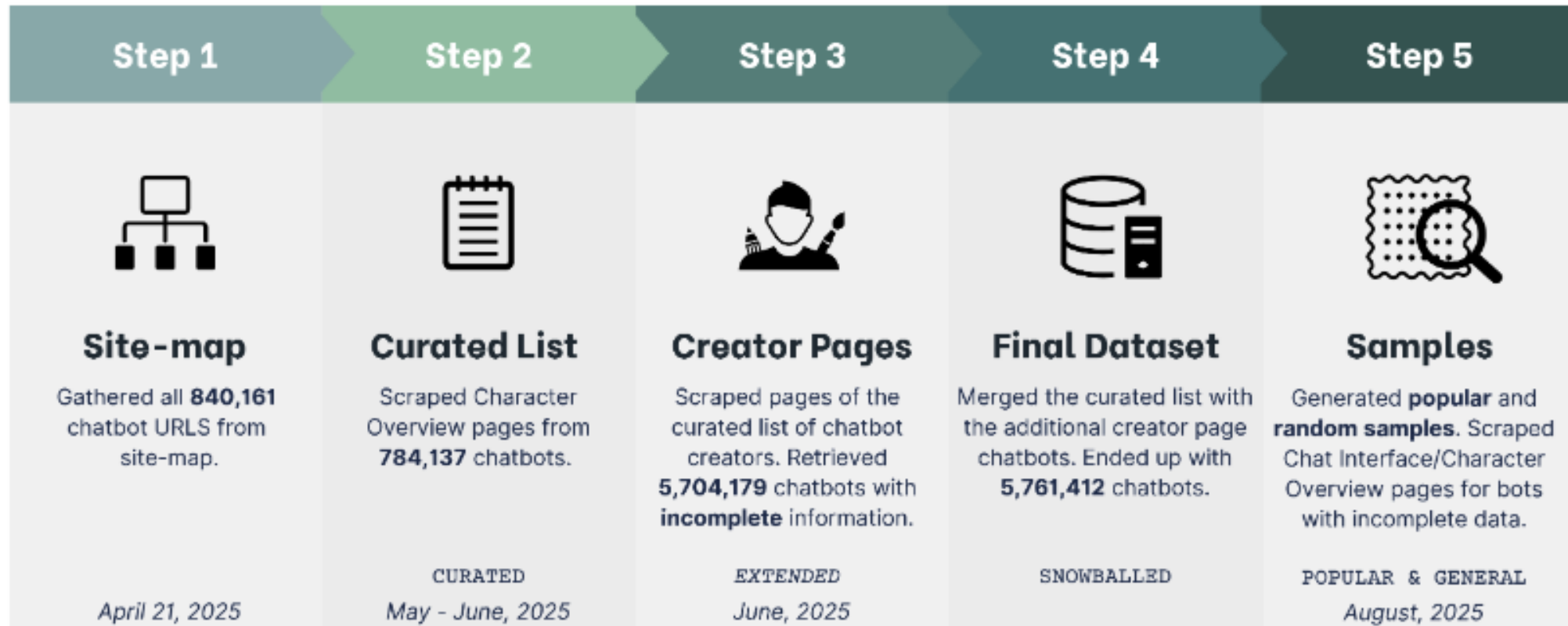
RQ1) What motivations do users share for creating and engaging with persona-based chatbots?

RQ2) What types of narratives are users interested in creating and exploring with characters?

- What types of relationship dynamics are embedded within them?

RQ3) What challenges do users express encountering when creating and interacting with persona-based chatbots?

Data collection pipeline



Results

Category	Subcategory	POPULAR (n=1468)	GENERAL (n=1000)
Relationship Type	General	32%	43%
	Intimate	63%	51%
	Non-Intimate	5%	6%
Relationship Quality	Power Dynamics	26%	16%
	Unhealthy Attachment	18%	9%
	Violent/Abusive	22%	12%

Table 3: Percentage of POPULAR and GENERAL cAI Descriptions that established specific relationship types and qualities for subsequent interaction.

Category	POPULAR (n=1468)	GENERAL (n=1000)
Fandom Characters	39%	57%
Organized Crime	8%	3%
School	20%	14%
High Stakes Situation	10%	10%

Table 4: Percentage of POPULAR and GENERAL cAI Descriptions that belonged to pre-existing universes or established specific contexts within which to interact.

Limitations of measurement studies

- Need to have a vantage point (access to the vantage point)
- Access barriers
 - Some sites don't let you crawl
- You have to be able to quantify what you're interested in
- No user experience data unless you run a mixed methods study
- Sometimes you collect the data, but don't find anything interesting

Activity:

You are a researcher tasked with improving how we detect scams online and helping users identify/recover from scams. Your team believes that scammers might be using generative AI tools to conduct their operations.

Task: What part of this problem could you solve with the tools we've covered, what research questions would you ask, and what would you need to do to collect data?

- Qualitative (Interviews)
- Surveys
- Usability
- Measurement (Web crawling/scraping)
- Put your responses here: <https://bit.ly/45RHBwH>

For next time:

- If you are new or haven't done it yet: Please sign up for Gradescope and Ed. Links are on the course website: <https://sociotechnical-security.github.io/>
- First set of readings and reading response due on Monday at 2 PM on Gradescope
- Start thinking about your semester project
- Office hours begin today, come say hi. 2nd floor of Coda.